

ID804

RGB LED Driver IC for Smart Ambient Light

General Description

This device is designed to intelligently respond to the sophisticated ambient lighting scenarios in evolving automotive interiors.

It incorporates a fully calibrated package containing three-color RGB LED chips and drive IC, ensuring precise calibration before shipment.

Through a single microcontroller, up to 4078 devices can be daisy-chained and controlled. Moreover, the microcontroller can detect the temperature and fault status of each device.

The addition of CAN transceivers between daisy-chained devices ensures seamless long-distance data communication.

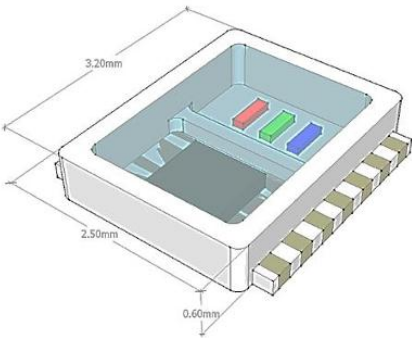
In an LED chain, each device is individually addressable via the serial bus. A proprietary bus protocol is used, optimized for this particular application.

Application

- Automotive: Backlight in Dashboard and Switch
- Automotive: Ambient lighting (Interior lighting)
- Functional lighting
- General lighting

Ordering Information

Product ID	Package	Body Size
ID804	DFN3225-14L	3.2x3.0x0.6mm



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1. Features:

- AEC-Q100 Grade 2 Qualified
- AEC-Q102-003 Qualified
- Max 4KHz, 12-bit PWM Dimming Resolution
- 8-bit brightness resolution for R/G/B LED
- Maximum of 4078 devices in one LED chain
- Bidirectional, half-duplex, 2Mbps, CRC protected serial communication
- White point calibration D65
- Support for 16 multicast address groups
- On-die oscillator
- Built-in diagnostic functions
- Auto addressing of LED via the serial bus in daisy chain
- Supply voltage 4.3V ~ 5.5V
- ESD, HBM protection (AEC Q100-002 ANSI/ESDA/JEDEC JS-001): $\pm 2\text{kV}$
- Qualified according to JEDEC MSL 2

2. Block Diagram

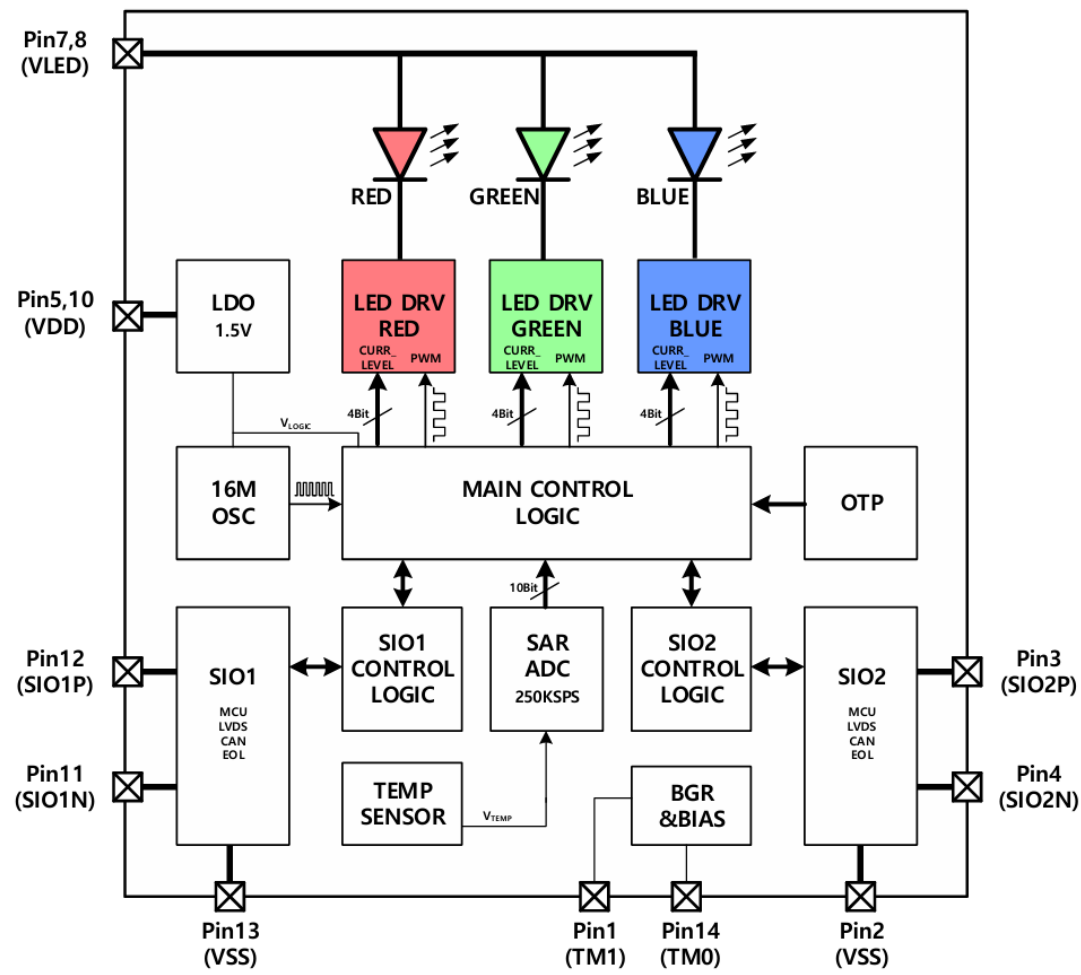


Fig 1. Functional Block Diagram

The device implements a communication for the reception of control commands and for providing device status and configuration data. Low side, configurable constant current sinks are provided for controlling 3 LEDs (i.g. RGB). The main control logic computes the PWM duty cycles from the incoming commands and applies the corresponding control values to the three PWM generators. The main control logic is also in charge of a periodic temperature monitoring and an appropriate duty cycle adjustment for the red PWM channel. The actual device temperature is obtained via an integrated analog-digital converter (ADC). The die temperature is updated in the corresponding registers, and the values can be read by the MCU by host commands. As each device is individually calibrated to compensate for LED variations, the corresponding parameters can be stored in an one-time-programmable memory (OTP). The calibration data stored in the OTP is loaded into the main control logic whenever the IC is turned on or reset command.

3. Pin Configuration

3.1. Pin Map

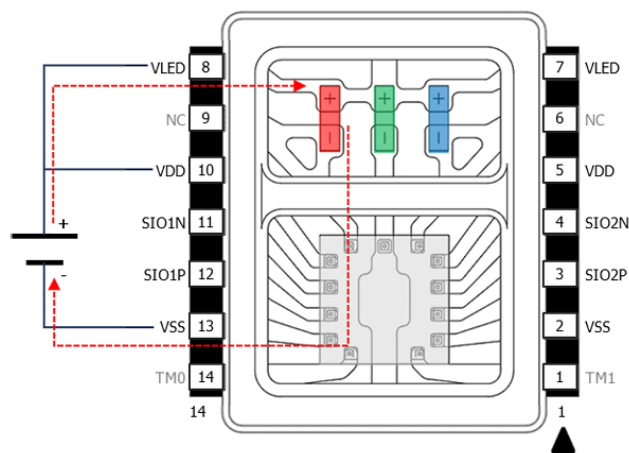


Fig 2. Top view with integrated LED and Driver IC

3.1. Pin Description

Pin No	Pin Name	Type	Description
1	TM1	IN	Connect to GND. (Test mode AMUX pin with a built-in pull-down resistor, 200KΩ)
2	VSS	GND	Power Ground
3	SIO2P	IN/OUT	Serial communication interface slave side, positive polarity
4	SIO2N	IN/OUT	Serial communication interface slave side, negative polarity
5	VDD	PWR	5V DC Supply voltage for IC
6	N.C	-	No connection (BLUE LED Cathode side)
7	VLED	PWR	5V DC Supply voltage for RGB LEDs
8	VLED	PWR	5V DC Supply voltage for RGB LEDs
9	N.C	-	No connection (RED LED Cathode side)
10	VDD	PWR	5V DC Supply voltage for IC
11	SIO1N	IN/OUT	Serial communication interface master side, negative polarity
12	SIO1P	IN/OUT	Serial communication interface master side, positive polarity
13	VSS	GND	Power Ground
14	TM0	I/O	Connect to GND (Test mode entry pin with a built-in pull-down resistor, 200KΩ)

Table 1. Pin Description

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4. General Product Characteristics

4.1. Absolute Maximum Rating

The absolute maximum ratings define values beyond which damage to the device may occur. Exposure to absolute maximum rating conditions for extended periods may affect device reliability. The functional operation of the device at these or any other conditions beyond the recommended operating ratings is not guaranteed.

Over operating free-air temperature range (unless otherwise noted) ⁽¹⁾

Parameter	Symbol	Min	Max	Unit
Input voltage range (with respect to GND unless specified)	V _{DD}	-0.3	7.0	V
	V _{LED}	-0.3	V _{DD} + 0.3	V
	TM0 / TM1	-0.3	V _{DD} + 0.3	V
Input/Output voltage range (with respect to GND unless specified)	SIO1P/N	-0.3	V _{DD} + 0.3	V
	SIO2P/N	-0.3	V _{DD} + 0.3	V
Junction to ambient thermal resistance	θj-a	90		°C/W
Soldering temperature	Reflow	255°C 30 sec Max.		°C
Storage temperature of IC	T _{STG}	-40	125	°C
Junction temperature of IC	T _J	-40	125	°C
ESD	HBM ⁽²⁾	±2000		V
	CDM ⁽³⁾	±500		V
	CDM(comer pin)	±750		V

Table 2. Absolute Maximum Ratings

- Note :
- (1) Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. These are stress ratings only, and do not imply functional operation of the device at these or any other conditions beyond those indicated under Recommended Operating Conditions. Exposure to absolute- maximum-rated conditions for extended periods may affect device reliability.
 - (2) AEC Q100-002 indicates that HBM stressing shall be in accordance with the ANSI/ESDA/JEDEC JS-001.
 - (3) AEC Q100-011

4.2. Recommended Operating Condition (V_{DD} =5.0V, T_A =25.0°C unless otherwise noted)

Parameter	Symbol	Min.	Typ.	Max.	Unit
DC supply voltage IC and LED	V _{DD} , V _{LED}	4.3	5	5.5	V
V _{DD} Ramp up time	T _{RAMP_VDD}	0.5			ms
Device ready for commands, Initialization time after V _{DD} > V _{POR} level	T _{INIT}			5.0	ms
Data rate for communication	F _{COMM}	1.8	2.0	2.2	Mbps
Ambient temperature of IC	T _{A_IC}	-40		105	°C
Ambient temperature of Multi Chip Module (include RGB LED chip)	T _{A_MCM}	-40		85	°C

Table 3. Recommended Operating Condition

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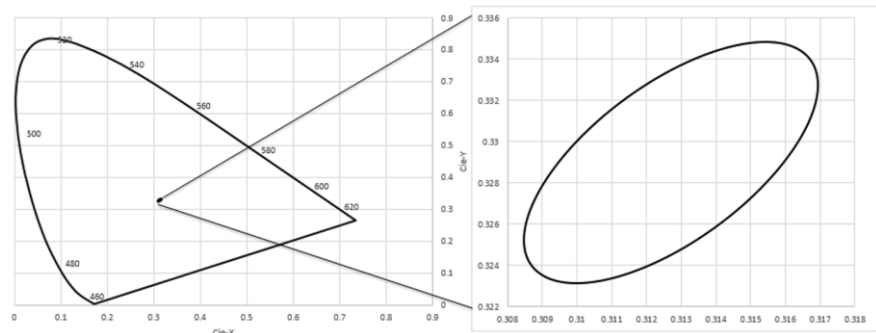
4.3. Electro-optical Characteristics (V_{DD}=5.0V, T_A=25.0°C , C_{VDD}=1μF, unless otherwise noted)

Parameter	Symbol	Color	Code	Conditions	Min.	Typ.	Max.	Unit
Dominant Wavelength ⁽¹⁾	λ _{D_R}	Red	W36	(255. 0, 0)	626	628	632	nm
	λ _{D_G}	Green	W25	(0, 255, 0)	528	530	534	nm
	λ _{D_B}	Blue	W12	(0, 0, 255)	458	462	464	nm
Luminous Intensity ⁽²⁾	I _{v_R}	Red		(255,0,0)		720		mcd
	I _{v_G}	Green		(0,255,0)		1920		mcd
	I _{v_B}	Blue		(0,0,255)		160		mcd
	I _{v_W}	White ⁽⁴⁾		(255,255,255)	-10%	2800	+10%	mcd
Forward Voltage	V _{F_R}	Red	D	@10mA		1.9		V
	V _{F_G}	Green	K	@10mA		2.4		V
	V _{F_B}	Blue	L	@10mA		2.6		V
View angle ⁽³⁾	φ	-		(255,255,255)		120		degree

Table 4. Electro-optical Characteristics

Note :

- (1) Dominant wavelength has an accuracy of ±2nm
- (2) Luminous Intensity is tested by a tester calibrated by CAS-140B(CIE LED_B) and has an accuracy of 10%
- (3) Viewing angle is the angle until 50% of brightness measured from the front part of LED.
- (4) D65(6500K), The above luminous intensity represents brightness at 100%.



Rank	Description	Center		Radius		Angle(°)
		x	y	a	b	
D65	Macadams 3steps	0.3128	0.3290	0.00669	0.00285	58.57

Table 5. Color Coordinates Group

4.4. Electrical Characteristics (V_{DD}=5.0V, T_A=30°C , C_{VDD}=1μF, unless otherwise noted)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
VDD BIAS SUPPLY						
V _{POR}	Power-On-Reset (POR)	VDD voltage for communication start			3.8	V
V _{uvth_hi}	Undervoltage Lockout threshold	VDD Rising, LED driver is enable UV_LVL[1:0] = 2b'11	4.10	4.25	4.40	V
V _{uvth_lo}		VDD Falling, LED driver is disable	3.80	4.05	4.30	V
V _{uvdet_hyst}	UVLO Detection hysteresis threshold	Guaranteed by design		160		mV
DC Supply Currents						
I _{Q_dpslp}	Deep sleep mode current at VDD	Deep Sleep mode, T _A = 30°C		590		uA
I _{Q_sleep}	Sleep mode current at VDD	Sleep mode, LED driver off, only IC		950	1050	uA
I _{Q_activ}	Active mode current at VDD	Active mode, LED driver on, only IC *		1.1	1.25	mA
*Total ID804 SiP current consumption shall include LED drive current. "(Typ 1.1mA+i(LEDs)/200)+i(LEDs)"						
RGB LED Driving current characteristic						
I _{O_LED_RED}	Driving current of Red LED	R_DATA[7:0]=0xFF, V _{OUT_R} =1V, R_CURR_MAX_LVL[3:0]=0xB	-15%	16.0	+15%	mA
I _{O_LED_GRN}	Driving current of Green LED	G_DATA[7:0]=0xFF, V _{OUT_G} =1V, G_CURR_MAX_LVL[3:0]=0xA	-15%	13.9	+15%	mA
I _{O_LED_BLU}	Driving current of Blue LED	B_DATA[7:0]=0xFF, V _{OUT_B} =1V, B_CURR_MAX_LVL[3:0]=0x5	-15%	7.8	+15%	mA
I _{O_LED_drift}	Constant output current aging drift	Guaranteed by design	-1%		+1%	%
I _{O_OFF_lkg}	Output leakage current	All LED is off, V _{OUTn} =5V			1	uA
Dimming parameter relationships between RGB parameters and PWM frequencies						
T _{SR_RS_LED}	LED current Rising time	I _{OUTn} (10% to 90%), Guaranteed by design			1	us
T _{SR_FL_LED}	LED current Falling time	I _{OUTn} (90% to 10%), Guaranteed by design			0.1	us
T _{DLY_ON_LED}	LED On command Propagation delay time	From end of command to 10% I _{OUTn} , Guaranteed by design		30	50	ns
T _{DLY_OFF_LED}	LED Off command Propagation delay time	From end of command to 90% I _{OUTn} , Guaranteed by design		30	50	ns

Table 6. Electrical Characteristics

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Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
Dimming Phase shift for RGB LED driving						
P _{SHIFT_Green}	Green Phase shift refer to PWM dimming freq	Guaranteed by design		0		degree
P _{SHIFT_Red}	Red Phase shift refer to PWM dimming freq	Guaranteed by design		90		degree
P _{SHIFT_Blue}	Blue Phase shift refer to PWM dimming freq	Guaranteed by design		270		degree
LED Fault diagnostic						
V _{TH_OPEN_hl}	LED Open diagnosis threshold voltage	LED Open, High		340		mV
V _{TH_OPEN_lo}		LED Open, Low		160		mV
V _{TH_Hys_OPEN}	LED Open hysteresis threshold voltage	Guaranteed by design		180		mV
V _{TH_SHORT}	LED Short diagnosis threshold voltage	SH_LVL[1:0] = 2b'10		4.4		V
T _{diag_filter}	LED fault diagnostic deglitch filter count	Guaranteed by design Counter clock is based on "F_PWM_DIV[3:0]" configuration		16		cnt
Main Oscillator						
Fosc_main	Internal Main Oscillator frequency		15.2 (-5%)	16	16.8 (5%)	MHz
Serial Interface						
f _{comm_data_input}	Allowed data rate for communication input		1.8 (-10%)	2.0	2.2 (10%)	MHz
f _{ME_data_sing}	Signal data rate for single ended Manchester code	Guaranteed by design Manchester coded=Valid data rate x 2			4.4	Mbaud
D _{comm_data_in}	Allowed duty cycle variance for communication input		45		55	%
t _{msg_interval}	Time between two messages	Guaranteed by design Send from MCU		6		us
Single-ended Physical layer parameters						
V _{IH,se}	input high level voltage		2.0		5.5	V
V _{IL,se}	input low level voltage		-0.3		0.7	V
R _{OUT_PU_SE}	Output driver pull-up resistance	I _{DRV_OUT} = -10mA		60	100	Ω
R _{OUT_PD_SE}	Output driver pull-down resistance	I _{DRV_OUT} = 10mA		60	100	Ω
T _{DLY_RXTX_SIN}	Propagation delay from RX to Tx on single-ended	Low to High			50	ns
		High to Low			50	ns

Table 6. Electrical Characteristics

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Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
LVDS Physical layer parameters						
V _{LVDS_comm}	Common mode voltage with active communication			1.2		V
V _{LVDS_off}	Common mode voltage with no active communication	Design info.		0		V
V _{LVDS_diff}	LVDS differential voltage	Design info.		300		mV
R _{LVDS_term}	LVDS termination resistance	Included IC	200		300	Ω
I _{LVDS_TX}	LVDS TX Current		1.35	1.50	1.65	mA
R _{SIO_Ppd}	SIO_P Pull down resistance	Included IC		260		kΩ
R _{SIO_Npd}	SIO_N Pull down resistance	Included IC		190		kΩ
TEMP Analog to Digital Converter						
ADC _{BIT}	Bit Resolution	Design info.		10		bit
F _{smp_rate}	Sampling Rate	Design info.		250K		SPS
SNDR	Signal to Noise and Distortion Ratio	Design info.	50	54		dBc
ERR _{conversion}	ADC conversion error	Design info.	-10		+10	LSB
ADC _{Troom}	ADC Value for Room Temp Sensing	T _A = 30°C, Design info. LED driver off, Register read		0x512		digit
ADC _{Thot}	ADC Value for Hot Temp Sensing	T _A = 120°C, Design info. LED driver off, Register read	TBD	0d897	TBD	digit

$$T_{j,mon} = 30^{\circ}\text{C} + \left(\frac{TEMPERATURE[9:0] - d512}{4.05}\right)$$

Table 6. Electrical Characteristics

4.5. Thermal Considerations

Need to suggest the maximum amount of power that the IC can consume under AEC-Q100 grade 2 conditions(T_A 105°C at still air). Thermal resistance characteristics would be variable depending on the IC, package, and PCB layer characteristics.

“ID804 of allowable IC power dissipation characteristics by ambient temperature”

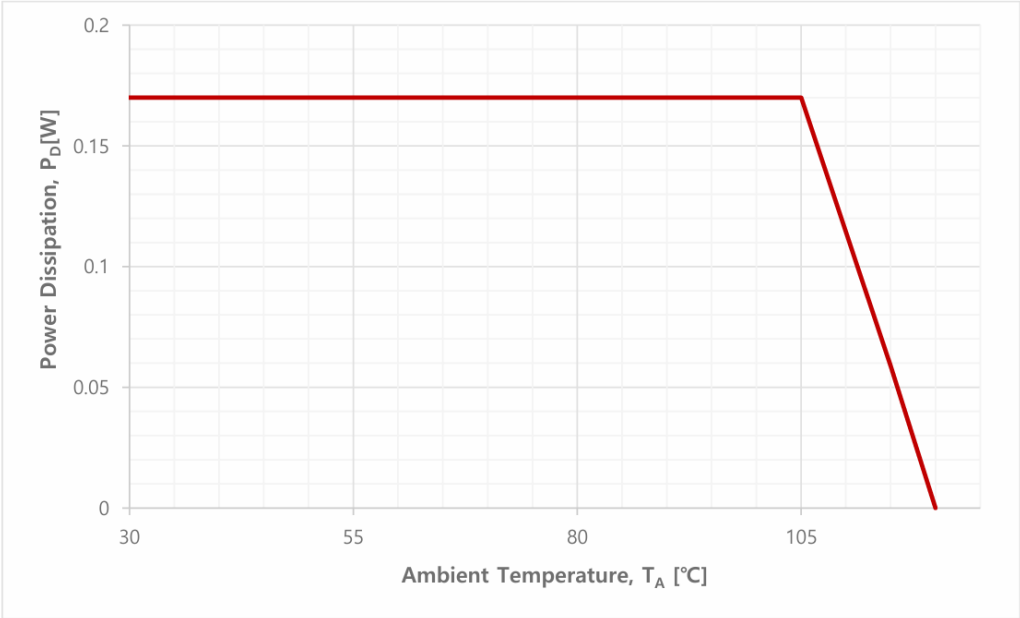


Fig 3. Thermal Considerations

5. Functional Description

5.1 Device States

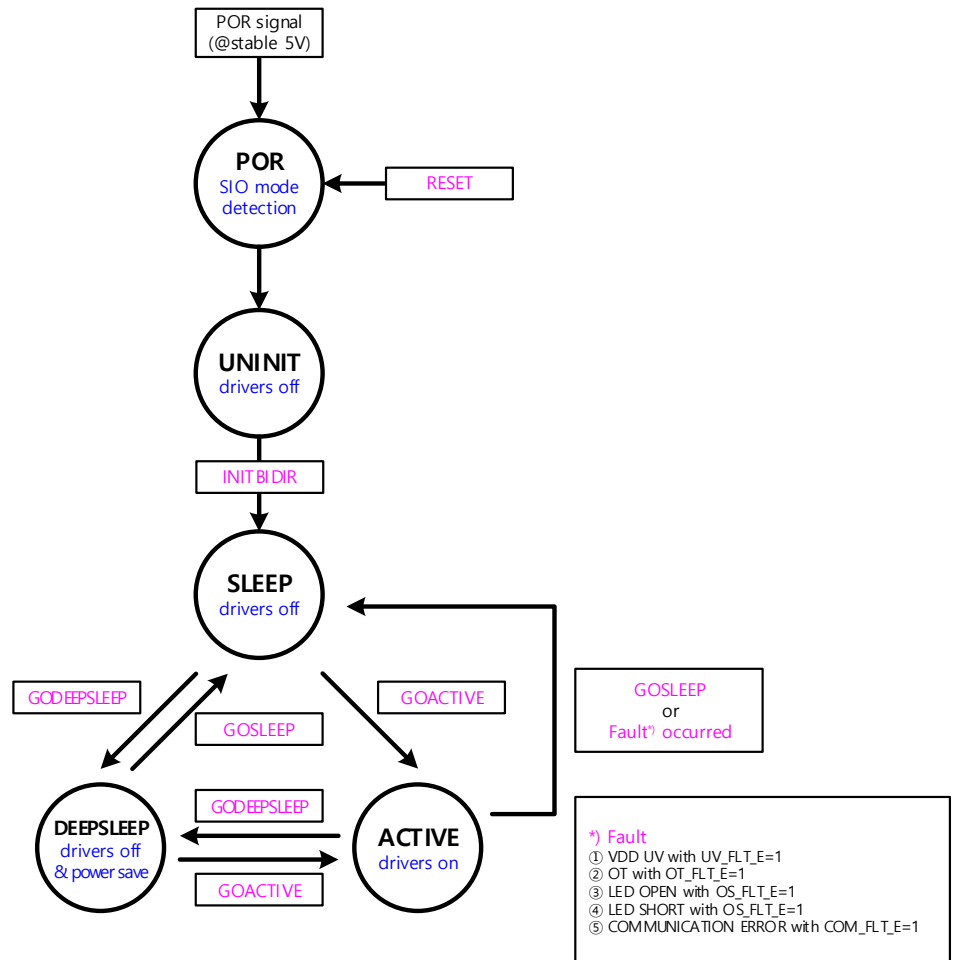


Fig 4. IC State

POR : The ID804 is in an off state.

UNINIT : The ID804 will reach this state automatically after power-up. PWM values are 0 and LED drivers disabled.

In this state, the ID804 only responds to a limited set of commands:

- INITBIDIR
- RESET (broadcast)
- CLRERROR (broadcast)
- SET_SETUP1 (broadcast)
- SET_SETUP2 (broadcast)
- SET_TEMPTH (broadcast)
- SET_TEMPHYS (broadcast)
- SET_RGB (broadcast)

Other telegrams are ignored and fault flag (STATUS2.COM_FLT) set. Broadcast telegrams are fast forwarded and addressed telegrams are forwarded after the telegram has been fully received.

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SLEEP

The ID804 enters SLEEP state after receiving an INITBIDIR command, after a failure event (if selected) or via GOSLEEP command.

In this state, the LED drivers are disabled but the last PWM parameters are stored, i.e., not reset to 0. An update of the PWM parameters is possible.

ACTIVE

The ID804 enters this mode after receiving the GOACTIVE command if no fault conditions are present. In this state the LED drivers are enabled with the respective current setting, PWM frequency and duty cycle for each channel. An update of the PWM parameters is possible.

After entering the ACTIVE state and LED drivers enabled, the open/short detection is executed once for every channel with a PWM value exceeding the minimum value. See Chapter 5-5.

DEEP SLEEP

The ID804 is in a power save mode with reduced current consumption. In this state, the ADC is disabled, and the LVDS receiver waits in a low power mode. When a signal is detected on the LVDS, the receiver switches to normal mode, transmits the data to the internal logic, and then returns to low power mode until the next LVDS signal is received.

5.2 Device Enable

When the supply voltage is connected and VDD increases above the V_{POR} level, or the RESET command is received, the ID804 undergoes the following startup sequence (Fig 5. Startup behavior of the ID804.

The example shows the MCU(left) and EOL(right) modes. After a RESET command, the device starts in the POR state.)

- 1. All registers are initialized to their default value (POR in Fig 5).
- 2. Trim settings are loaded and checked for integrity. It takes max 120us. If a mismatch is detected, the OTPCRC flag in the STATUS2 register is set.
- 3. The voltage levels at each SIOx pin are checked to detect the communication modes (shaded signals in Fig 5).
- 4. The ID804 enters the UNINIT mode and is now ready to receive the first command (UNINIT in Fig 5).

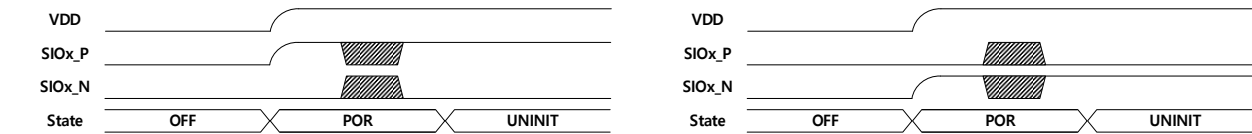


Fig 5. Startup behavior of the ID804. The example shows the MCU (left) and EOL (right) modes. After a RESET command, the device starts in the PoR state.

Note : The whole sequence takes max. 5 ms.
(When sequence is started caused by RESET command, the The whole sequence takes approximately 1 ms.)

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5.3 Communication

5.3.1 Single Ended, LVDS Link and Data Structure

5.3.1.1 Overview

- Bi-directional
- LVDS - differential Manchester coded
- Single ended – Manchester coded & Manchester decoded or NRZ(clock & data) Push-pull interface, clock is invertible
- Communication direction and SIOx connection is flexible
- Communication speed is typ. 2Mbps Manchester coded

5.3.1.2 Communication modes

Each one of the 2 available SIOx interfaces do support 4 different modes.

Each SIOx only can operate in semi-duplex mode meaning it either does receive or transmit a telegram at the same time. However, it is possible that a device does receive a telegram at one SIOx and start transmitting at the other SIOx interface.

The 4 communication modes are:

1.“LVDS-mode”:

The SIOx communication is using LVDS signaling for transmitting and receiving a communication telegram. The bit stream is using Manchester encoding to transfer DATA & CLK on the same line.

2.“MCU-mode”:

In this mode a SIOx is using single-ended mode for the signaling. Received telegrams are Manchester encoded on SIOx_P port. For transmitting, the device is decoding the Manchester signal. SIOx_P will deliver the DATA in NRZ format and SIOx_N delivers the corresponding clock.

3.“EOL(End Of Line)-mode”:

SIOx port does consider itself to be at the end of the daisy chain.

4.“CAN-mode”:

In this mode, it is possible to communicate via a CAN FD transceiver. SIOx_P is always in output state and transmits telegram as Manchester encoded signals (single-ended), SIOx_N is always in input state and receives telegrams as Manchester encoded signal (single-ended)

5.3.2 Communication mode selection

Communication mode auto detection takes place during initialization phase via voltage level check on SIOx_P & SIOx_N and this mode cannot be changed during the operation cycle. After a HW Reset (POR) or RESET command received the Initialization with mode selection take place again

Mode	Pin Configuration		I/O		Signal Type	Data Encoding	
	SIOx_P	SIOx_N	SIOx_P	SIOx_N		TX	RX
MCU	Pull-Up	Pull-Down	Bi-direction	Output or Hi-Z	Single-ended	NRZ (CLK & Data)	Manchester
LVDS	Pull-Down (By Internal Resistor)	Pull-Up	Bi-direction	Bi-direction	LVDS	Manchester	Manchester
CAN	Pull-Up	Pull-Up	Output	Input	Single-ended	Manchester	Manchester
EOL	Pull-Down	Pull-Up	-	-	-	-	-

Table 7. Communication mode selection

5.3.3 MCU-mode communication physical layer and bit coding

Single ended communication in the master to slave direction is performed with so-called Manchester encoded. In that case a bit stream is sent via SIOx_P only; SIOx_N of the device is ignored. Slave to master communication is performed with NRZ coding. In that case SIOx_N provides a clock signal. The clock signal is inverted if it's selected via setup register. The typical communication data rate is 2Mbps.

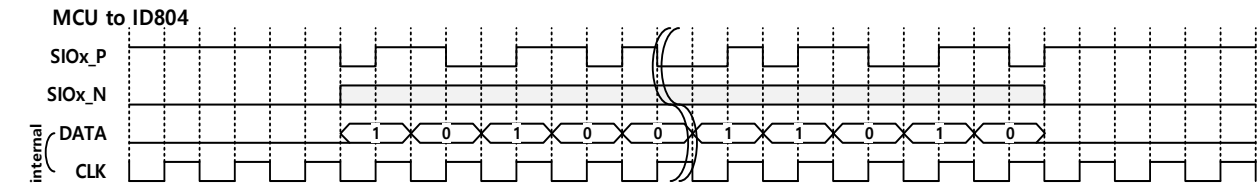


Fig 6. Example of MCU to ID804(RX) communication

The receiving clock could be inverted via setup register (SETUP2.CLK_INV_E)

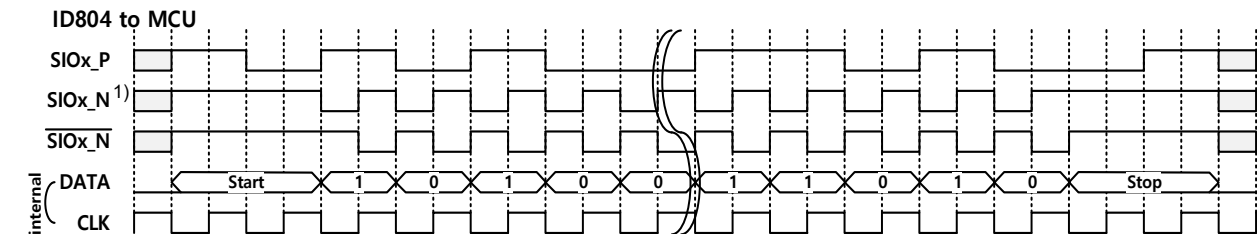


Fig 7 Example of ID804 to MCU (TX) communication

Note:

- 1. The receiving clock could be inverted via setup register (SETUP2.CLK_INV_E)

5.3.4 LVDS-mode communication physical layer and bit coding

For differential communication the termination resistance R_{LVDS_term} is 200 Ohm, differential voltage V_{LVDS_diff} is typ. 300 mV, data rate is typ. 2 Mbps. Common mode voltage V_{LVDS_comm} is typ.1.2 V, it means the max voltage on the communication lines during communication is typ. 1.35 V and the min. voltage is typ. 1.05 V

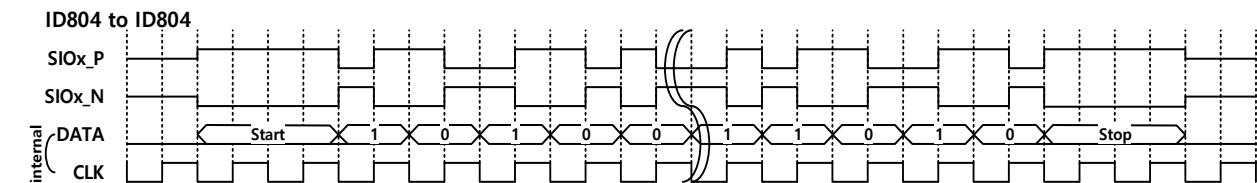


Fig 8. Example of ID804 to ID804 communication

5.3.5 CAN-mode communication physical layer and bit coding

If a transceiver is detected on SIOx the output and input message is via single ended & Manchester encoded communication with typ. 2 Mbps data transfer rate. For the transceiver the data transfer rate depends on the data signal due to the Manchester encoded data. Means up to the double transfer rate is needed (4 Mbps). Therefore, the use of a 5 Mbps CAN-FD transceiver is required.

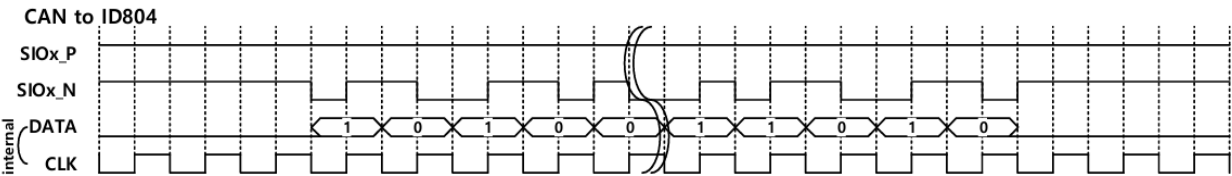


Fig 9 Example of CAN Transceiver to ID804 communication

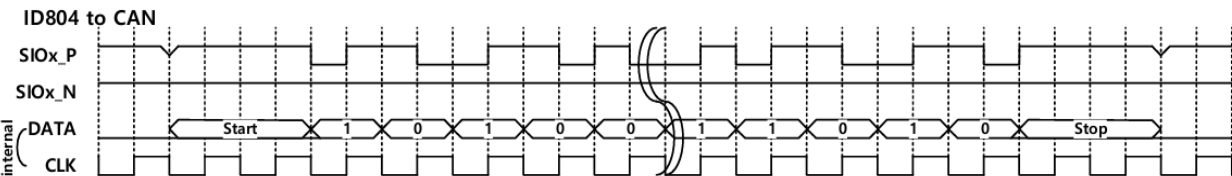


Fig 10 Example of ID804 to CAN Transceiver communication

5.3.6 Message Frame Format

ID804 has 3 types of frame format according to DATA bit. If the data bits are 12 or 24 bits, the upstream and downstream communication formats are the same. The 0-bit data frame format [Figure 13] can only be used for downstream communication.

Byte 1				Byte 2				Byte 3				Byte 4				Byte 5				Byte 6				Byte 7																															
55	54	53	52	51	50	49	48	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PREAMBLE				DEV_ADDR [11:0]								COMMAND [7:0]				DATA [23:0]																														CRC [7:0]									

Fig 11. 24 bit Data Frame Format

Byte 1								Byte 2								Byte 3								Byte 4								Byte 5								Byte 6							
47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PREAMBLE								DEV_ADDR [11:0]								COMMAND [7:0]								DATA [11:0]								DUMMY [3:0]								CRC [7:0]							

Fig 12. 12 bit Data Frame Format

Byte 1								Byte 2								Byte 3								Byte 4							
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PREAMBLE				DEV_ADDR [11:0]								COMMAND [7:0]								CRC [7:0]											

Fig 13. 0 bit Data Frame Format

- The bit order is MSB first.
- PREAMBLE (4 bit) : Fixed to 4'b1010.
Avoid messages with incorrect preamble as this could lead to forwarding of garbage telegrams along the chain and to raising the STATUS2.COM_FLT.
 - DEV_ADDR (12 bit) : Allowed values for downstream messages are 0x000 (broadcast) and 0x001 to 0xFEE (unicast nodes) and 0xFEF to 0xFFE (multicast). Upstream messages always contain the device address of the sender in the DEV_ADDR field. Not all commands are allowed in a broadcast message. See Chapter 5-6
 - COMMAND (8 bit) : See Chapter 5-6
 - DATA (24bit or 12bit or 0bit) : DATA field has bits according to COMMAND. See Chapter 5-6

- DUMMY (4bit) : In case of 12bit Data Frame Format, 4bit DUMMY data (4'b1010) inserted to match the data size to BYTE.
- CRC (8 bit) : See Chapter Cyclic Redundancy Check

5.3.7 Message Handling

5.3.7.1 Broadcast

An address of 0x000 initiates a broadcast, i.e., the same message is sent and received by all nodes on the chain. Whether a command can be broadcast is detailed in Chapter 5-6.

Broadcast Read or SR command is first received by all devices via the frame in downstream direction. The last node in the chain then immediately transmits its response frame upstream. The response frame's data field depends on the actual command. The response frame's address field is set according to the own device's address. All the nodes upstream forward all received response frames until a frame with the address of their adjacent node is received. Then the respective node transmits its own response frame. This procedure lasts until the chain's first node has transmitted its response frame.



Fig 14. Example of broadcast write

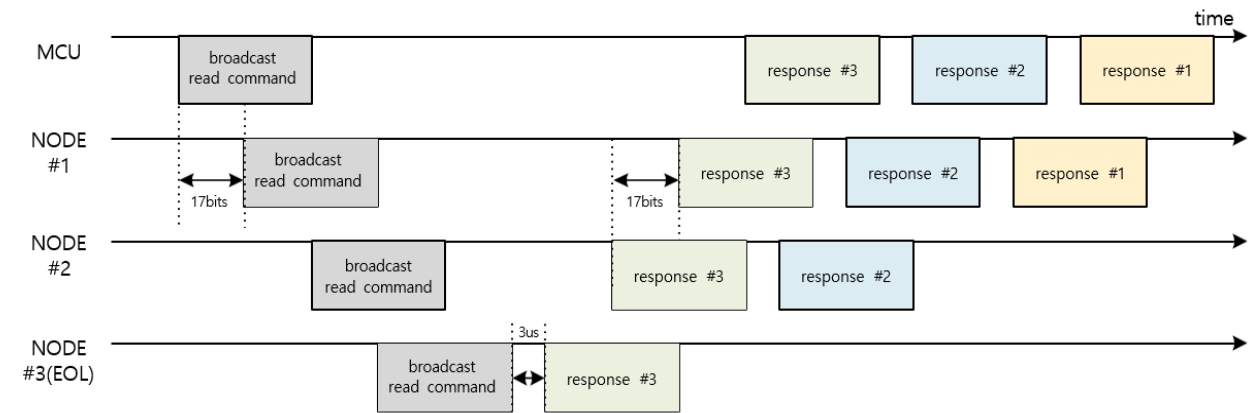


Fig 15. Example of broadcast read(or SR)

5.3.7.2 Multicast

An address of 0xFE0 to 0xFFE initiates a multicast, i.e., the same message is sent and received by same defined group nodes on the chain. Multicast group can be defined by MCAST command. Whether a command can be multicast is detailed in Chapter 5-6.

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5.3.7.3 Forwarding and Fast Forwarding

Message forwarding refers to the direct forwarding of a message without modification. Messages are forwarded to the next device if the address in the message frame is the broadcast address, or multicast address, or it is different from the node’s address.

If a message is to be forwarded, forwarding starts exactly one cycle after the preamble and address field are fully received, i.e., after 17 bits. This is referred to as fast-forwarding. When a node is uninitialized, broadcast messages are fast-forwarded and multicast(0xFE0 to 0xFE7) & unicast(0x001 to 0xFE7) & abnormal(0xFFFF) addressed messages are forwarded only after the full telegram has been received.

5.3.7.4 Responses and SR (Status Requests)

Some commands trigger a response from the addressed node. In this case, the address field contains the address of the sending node to indicate the origin of the message. Once the node receives the end of the command frame, the corresponding response will occur approximately 2us (in case of broadcast) or 26us (in case of unicast) later.

Some commands do not send a response but allow a SR(Status Request). The SR(Status Request) is done by specific command. See Chapter 5-6. In case of broadcast response or SR see Chapter Broadcast.

5.3.7.5 Waiting Time between Telegrams

ID804 is not accepting a new telegram while a telegram is being received or transmitted on the other port. Likewise, simultaneous upstream and downstream messages are not supported.

Therefore, a waiting time must be added by the MCU between sending two telegrams in order to ensure that all receiving and transmitting activities have been concluded.

The minimum waiting time between two downstream telegrams should be 6 μs (edge to edge) and applies to write-only or without response request commands. If an answer has been requested, the next telegram should only be sent after the response has been received by the MCU.

5.3.7.6 Communication Errors

The ID804 requires certain communication errors to be detected by each device in a daisy chain.

The below Table 9 summarizes all relevant error conditions.

IC State	Preamble	Address	Command	CRC ^{*)}	Non Target DEV		Target DEV	
					COM_FLT flag	Data Forwarding	COM_FLT flag	Data Forwarding
Uninitialized state (UNINIT)	Incorrect	-	-	-	Set	Stop	Set	Stop
	Correct	Unicast	-	-	Set	No stop	-	-
		Multicast	-	-	Set	No stop	-	-
		Broadcast	Incorrect	-	-	-	Set	No stop
			Correct	Incorrect	-	-	Set	No stop
Initialized State (SLEEP, NORMAL, DEEPSLEEP)	Incorrect	-	-	-	Set	Stop	Set	Stop
	Correct	Unicast	Incorrect	-	No set	No stop	Set	Stop
			Correct	Incorrect	No set	No stop	Set	Stop
		Multicast	Incorrect	-	No set	No stop	Set	No stop
			Correct	Incorrect	No set	No stop	Set	No stop
		Broadcast	Incorrect	-	-	-	Set	No stop
			Correct	Incorrect	-	-	Set	No stop

Table 8. Communication Errors Table

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The ID804 performs some error detection even for messages that are to be forwarded (CRC). If an error is detected in such a case, the COM_FLT flag is set and the message is still forwarded, i.e., communication is not stopped.

In UNINIT state(Uninitialized state), only broadcast address(0x000) and limit set of commands can accept "Correct" message. Other address (0x001 ~ 0xFFFF) messages are ignored. And still forwarded and set COM_FLT flag.

Errors due to incorrect command is only detected if the device is addressed by unicast or multicast or broadcast.

5.3.7.7 Communication Timeout

The READ commands or SR WRITE commands trigger a response from the addressed or all nodes. Each node needs to await all responses originating from the nodes downstream. Thereafter either the node's own response is transmitted or new commands are accepted. If new command(downstream) is received during response waiting, it is ignored and COM_FLT flag set on received node. Only the last node in the LED chain may immediately transmit its response. In case there is an error with the chain, not all expected responses may arrive. Thus, each of the commands expecting a response waits for a certain time only and then returns to its previous state without having transmitted the node's response data. The lengths of the timeouts depend on register setting command (SET_TIMEOUT). They are calculated to account for the worst-case oscillator frequency tolerance. I.e. the waiting node has a high-speed clock and all the nodes waited for have a low speed clock. The hardware implementation uses an internal clock for the timeout counter. The resolution is internal clock($F_{osc_main}(Typ. 16MHz) * (2^{10})$).

5.3.7.8 Cyclic Redundancy Check

The CRC field contains the CRC8 check word calculated over the full message excluding the CRC field.

CRC Implementation: $x^8 + x^5 + x^3 + x^2 + x + 1$.

Polynomial: 0x2F (normal notation)

Initial value: 0x00

XOR value: 0x00

Direction: MSB first

Note that this implementation uses the same polynomial but is not compatible with the AUTOSAR implementation because of the different initial and XOR values of 0xFF.

Note: The calculation direction for the CRC field in the non-volatile memory is LSB first.

5.4 Color Control and Temperature Monitor

5.4.1 PWM Generation

The ID804 incorporates three independent PWM channels, one for each LED. The PWM dimming resolution internally controlled by the IC is 12bit. The supported duty cycles are 0/4096 to 4095/4096. The PWM output frequency is set by SETUP1.F_PWM_DIV register setting. The frequency is decided by $(F_{osc_main}(Typ. 16MHz)) / ((F_PWM_DIV + 1) \times 4096)$. The PWM frequency is reduced to the quarter of set frequency when RGB data $\leq$ decimal 3 and the PWM frequency is reduced to the half of set frequency when decimal $4 \leq$ RGB data $\leq$ decimal 7. The output frequency is determined independently for each of the PWM channels.

5.4.1.1 PWM Update

When a new PWM duty cycle must be applied, this is always done at the end of a PWM duty cycle. I.e. the PWM always completes an output cycle using the previously active-duty cycle and starts the next output cycle using the updated duty cycle.

5.4.1.2 Phase Shift

To spread the current consumption of the LEDs over time, a phase shift enabled between the three PWM channels. It retains even if the output frequency of the channels is different. The fixed phase shift is defined in the following Table 10. Please note the absolute phase shift times are nominal values. I.e. they are subject to vary with the internal oscillator's frequency.

PWM Channel	Related Phase Shift	Unit
Green	0	Degree
Red	90	Degree
Blue	270	Degree

Table 9. PWM Phase Shift

5.4.2 Brightness Control Scaler

In operation, the color brightness of each LED is controlled via an 8-bit RGB register. This provides a color space of 3 x 8 bit = 24 bit. The 8-bit RGB value is scaled up to the 12-bit value using by Brightness calibration value(PWM max value). The PWM max values is stored OTP. The scaling equations of the red and green and blue channel are given in Table 11.

Color	Scaling equation
Red	$PWM_{RED} = (PWM\ max \times R_data + 128) \gg 8$
Green	$PWM_{GREEN} = (PWM\ max \times G_data + 128) \gg 8$
Blue	$PWM_{BLUE} = (PWM\ max \times B_data + 128) \gg 8$

Table 10. Scaling equation

5.4.3 Temperature Monitor

For temperature monitoring, the ID804 has a on-chip analog temperature sensor. This can be measured by ADC and value can be read from TEMPERATURE register through READ_TEMP command. This value allows monitoring the temperature of the device and compensating the brightness of the integrated LED based on temperature.

Parameter	Description	Equation	Unit
$T_{J, det}$	Junction temperature monitor	$T_{j, mon} = 30^{\circ}\text{C} + \left(\frac{TEMPERATURE[9:0] - d512}{4.05} \right)$	$^{\circ}\text{C}$
TEMPERATURE [9:0]	Junction Temperature value converted by ADC	10 bits	digit

Table 11. Temperature equation

5.5 Diagnostics

The ID804 provides built-in diagnostic functions for detection of fault conditions.

If a fault is detected, the respective fault flag bit is set. The fault flag bit can only be cleared when the STATUS2 register has been read by the MCU (either directly or via status request), all flag bits have been cleared using CLRERROR, or the device has received a RESET command.

If configured, the ID804 will automatically turn off the LED drivers when a failure is detected and change to SLEEP state to prevent damage to the part. This helps protect against device failures, ensuring communication is always possible throughout the daisy chain. See Chapter 5.6.2.1 for more details.

- **Overttemperature(OT)** : This fault occurs when the IC junction temperature rises above the overtemperature threshold for more than 5 consecutive EVENT_CYC cycles time. It can only be cleared the temperature has fallen below the overtemperature release threshold.
See Chapter 5.6.2.5 for details on controlling the threshold values.
- **Undervoltage (UV)**: This fault occurs when the supply voltage(VDD) of the ID804 drops below the undervoltage threshold for more than 5 consecutive EVENT_CYC cycles time.
Refer to the Chapter 5.6.2.2 for more details.
- **LED open / short**: This fault occurs when the LED driver pin voltage remains below the LED-open threshold voltage or above the LED-short threshold voltage for 16 consecutive FPWM_DIV cycles time. These fault conditions are tested individually for each channel, but only after the GOACTIVE command has been received and the PWM value of the channel is set to a value higher than the minimum value. Refer to the Chapter 4.4 for more details.
- **Communication Error** : See Chapter 5.3.7.6
- **Communication Timeout** : See Chapter 5.3.7.7

5.5.1 Fault Reaction

Fault	Detection	Fault bit	Configuration	Reaction
VDD UV	$VDD < V_{uvth_lo}$	UV_FLT	UV_FLT_E=0	Only Fault bit set
			UV_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
OT	$T_J > \text{Temperature threshold}$	OT_FLT	OT_FLT_E=0	Only Fault bit set
			OT_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
LED OPEN	$V_{LED} < V_{TH_OPEN}$	OPEN_R/G/B	OT_FLT_E=0	Only Fault bit set
			OT_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
LED SHORT	$V_{LED} < V_{TH_SHORT}$	SHORT_R/G/B	OT_FLT_E=0	Only Fault bit set
			OT_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
COMMUNICATION ERROR	See Table 8	COM_FLT	COM_FLT_E=0	Only Fault bit set
			COM_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
COMMUNICATION Timeout	See 5.3.7.7	T_OUT		Only Fault bit set

Table 12. Fault Reaction Table

5.6 ID804 Commands

The ID804 supports the following commands..

Command	Normal Command			SR (Status Request) Command			Data bits
	Encoding	Broadcast possible	Multicast possible	Encoding with SR	Broadcast possible	Multicast possible	
RESET	0xB1	O	O		-	-	0
INITBIDIR	0xB2	X	X		-	-	0
CLRERROR	0xB3	O	O	0xF3	O	X	0
GOSLEEP	0xB4	O	O	0xF4	O	X	0
GOACTIVE	0xB8	O	O	0xF8	O	X	0
GODEEPSLEEP	0xBC	O	O	0xFC	O	X	0
SET_SETUP1	0x86	O	O	0xC6	O	X	12
SET_SETUP2	0x87	O	O	0xC7	O	X	12
SET_MCAST	0x88	X	X	0xC8	X	X	24
SET_RGB	0xA0	O	O	0xE0	O	X	24
SET_TEMPTH	0x8A	O	O	0xCA	O	X	12
SET_TEMPYS	0x8B	O	O	0xCB	O	X	12
SET_CURR_MAX_LVL	0x8F	O	O	0xCF	O	X	12
SET_TIMEOUT	0x9A	O	O	0xDA	O	X	12
READ_STATUS1	0x41	O	X	-	-	-	12
READ_STATUS2	0x42	O	X	-	-	-	12
READ_TEMP	0x43	O	X	-	-	-	12
READ_TEMPST	0x44	O	X	-	-	-	24
READ_RESERVED	0x45	O	X	-	-	-	12
READ_SETUP1	0x46	O	X	-	-	-	12
READ_SETUP2	0x47	O	X	-	-	-	12
READ_MCAST	0x48	O	X	-	-	-	24
READ_RGB	0x60	O	X	-	-	-	24
READ_TEMPTH	0x4A	O	X	-	-	-	12
READ_TEMPYS	0x4B	O	X	-	-	-	12
READ_PWM_RED_VAL	0x4C	O	X	-	-	-	12
READ_PWM_GREEN_VAL	0x4D	O	X	-	-	-	12
READ_PWM_BLUE_VAL	0x4E	O	X	-	-	-	12
READ_CURR_MAX_LVL	0x4F	O	X	-	-	-	12
READ_TIMEOUT	0x5A	O	X	-	-	-	12

Table 13. ID804 Commands

If the ID804 receives a command that is not listed above, a communication error is raised.

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5.6.1 State control command

5.6.1.1 Reset

RESET:

Performs a complete reset of one or all devices. The effect is identical to a power cycle (see Chapter 5.1). All register values are set to their default values, all error flags are cleared, the communication mode detection is restarted, LED drivers are turned off, and the address is set to 0xFFFF. The device enters the UNINIT state.

After issuing the RESET command, wait for at least 1ms before sending out the next command. This command has a special characteristic for more easily re-initialize the chain, in case of some ICs have a reset by fault in the chain.

5.6.1.2 Initialize Bi-Directions

INITBIDIR:

Initiates the automatic addressing of the chain. The command shall be addressed to the first device in the chain always with address 0x001. The last device in the chain (indicated by the EOL mode, see Chapter 5.3.1.2) returns its address to the MCU. Subsequent commands should only be sent after the response has been received by the MCU.

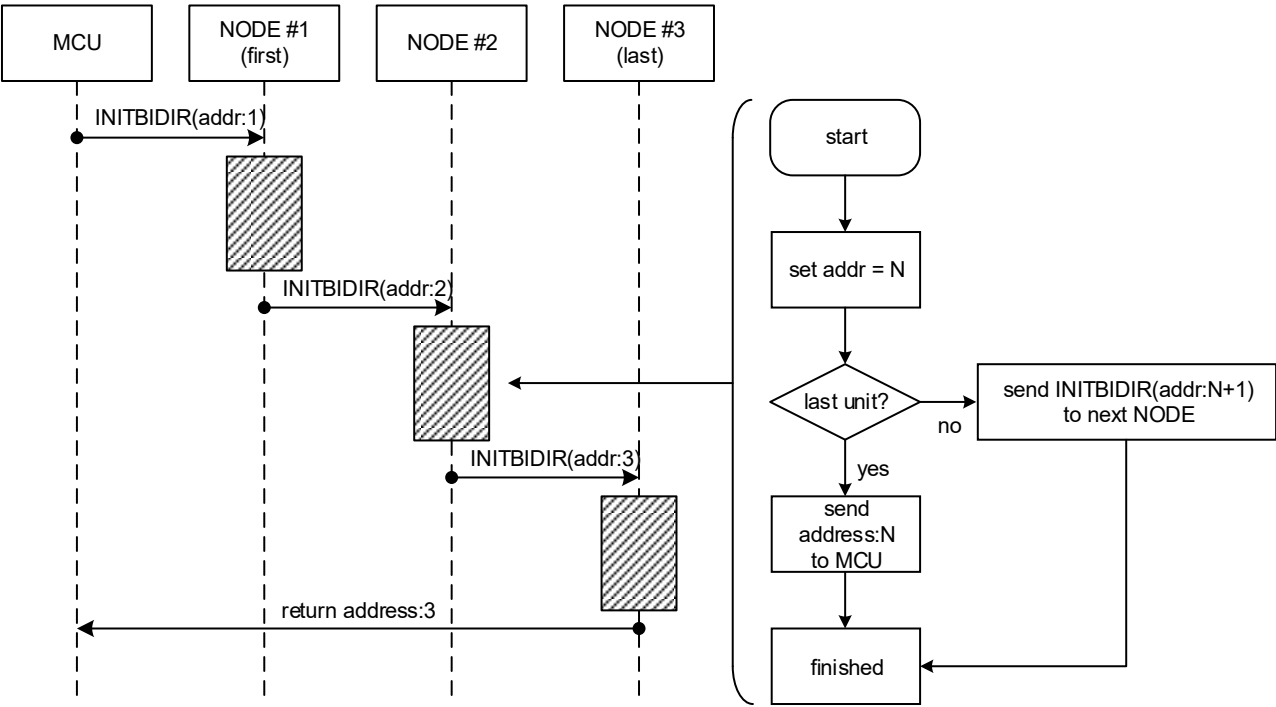


Fig 16. Automatic addressing feature

If the maximum number of node addresses has been reached (0xFEE), the device stops the initialization procedure and returns its address (0xFEE) to the MCU.

If a node is already initialized, the command is ignored but the telegram is forwarded with incremented address and the last device returns its address. This can be used to selectively initialize individual devices in the chain after a RESET or POR. In this case, do not initialize the chain with an address different from 0x001 as this could cause undefined behavior.

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Note that, when the last device is not in EOL mode (e.g., CAN-FD), it will not recognize itself as last device. Then, the application needs to ensure that the chain length is correctly received by the MCU.

Uninitialized nodes interpret this command.

Note : This command is not allowed as broadcast. If the broadcast address or multicast address is used nevertheless, all devices in the chain will be initialized to address 0x000 and no response telegram will be sent. Proper operation can be restored with a broadcast RESET followed by INITBIDIR with address 0x001

Note : If the command is sent using the broadcast address 0x000 into an already addressed chain, all devices will set the COM_FLT and the last device will return the TEMP & STATUS2 register with address 0x000.

5.6.1.3 Clear Error flags

CLRERROR:

Clears all error flags in one or all devices. If an error condition still persists, the error flag is automatically set again.

If the CLRERROR with SR command received, the device returns the TEMP & STATUS2 register after the error flags have been cleared.

5.6.1.4 Change state to SLEEP

GOSLEEP :

Sends one or all devices into SLEEP state. See Chapter 5.1

5.6.1.5 Change state to ACTIVE

GOACTIVE :

Sends one or all devices into ACTIVE state. See Chapter 5.1

5.6.1.6 Change state to DEEP-SLEEP

GODEEPSLEEP :

Sends one or all devices into DEEPSLEEP state. Chapter 5.1

5.6.2 Device Setup command

5.6.2.1 Setup for Device setting #1

SET_SETUP1 / READ_SETUP1 :

Write / Return the SETUP1 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	RSVD	OS_FLT_E	OT_FLT_E	UV_FLT_E	COM_FLT_E	RSVD	PH_SHIFT_E	LG_E	F_PWM_DIV[3:0]			
Default	0	0	0	0	0	1	0	0	0x0			
Access	RW	RW	RW	RW	RW	RO	RW	RW	RW			
RSVD			0: Default.									
OS_FLT_E			0: LED open/short fault flag, OS_FLT, is raised. 1: OS_FLT bit is raised, and LED drivers are disabled.									
OT_FLT_E			0: Overtemperature fault flag, OT_FLT, is raised. 1: OT_FLT bit is raised, and LED drivers are disabled.									
UV_FLT_E			0: Undervoltage fault flag, UV_FLT, is raised. 1: UV_FLT bit is raised, and LED drivers are disabled.									
COM_FLT_E			0: Communication fault flag, COM_FLT, is raised. 1: COM_FLT bit is raised, and LED drivers are disabled.									
RSVD			1: Read Only									
PH_SHIFT_E			0: No shift 1: G starts driving the sink-output at 0 degree, and then R, 90 degree and B, 270 degree, respectively.									
LG_E			0: Low Grayscale PWM disable. 1: Low Grayscale PWM enable. If R/G/B each 8-bit data ≤ 0x3, the PWM frequency is FPWM/4, and if 0x4 ≤ data ≤ 0x7, the PWM frequency is FPWM/2.									
F_PWM_DIV[3:0]			PWM frequency divider. The PWM Frequency is activated in only active mode. FPWM = (mclk/4096) / (F_PWM_DIV+1)									

5.6.2.2 Setup for Device setting #2

SET_SETUP2 / READ_SETUP2 :

Write / Return the SETUP1 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	RSVD	RSVD	RSVD	RSVD	RSVD	CLK_INV_E	RSVD	EVENT_CYC	UV_LVL[1:0]		SH_LVL[1:0]	
Default	0	0	0	0	0	0	0	0	0x11		0x10	
Access	RO	RO	RO	RO	RO	RW	RO	RW	RW		RW	
CLK_INV_E		CLK polarity for MCU mode. 0: Inverted disabled. 1: Inverted enabled.										
RSVD		0: Read Only										
EVENT_CYC		This bit determines a period to detect fault events or to update the ADC output value. EVENT_CYC period & ADC_SEQ_CLK period 0: 31.25KHz (typical 32us) & 125KHz (typical 8us) 1: 62.5KHz (typical 16us) & 250KHz (typical 4us)										
UV_LVL[1:0]		VDD UVLO detection level. 2'b00 : VUV_TH_Rise = 3.65V, VUV_TH_Fall = 3.50V 2'b01 : VUV_TH_Rise = 3.80V, VUV_TH_Fall = 3.65V 2'b10 : VUV_TH_Rise = 4.00V, VUV_TH_Fall = 3.85V 2'b11 : VUV_TH_Rise = 4.25V, VUV_TH_Fall = 4.05V (default)										
SH_LVL[1:0]		LED Short detection level. 2'b00 : VS_TH = 3.95V. 2'b01 : VS_TH = 4.20V. 2'b10 : VS_TH = 4.40V. (default) 2'b11 : VS_TH = 4.70V.										

5.6.2.3 Setting of Multicast group Address

SET_MCAST / READ_MCAST :

Write / Return the MCAST register

Bit	[23]	[22]	[21]	[20]	[19]	[18]	[17]	[16]	[15]	[14]	[13]	[12]
Name	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	MCAST_ADDR [15:12]			
Default	0	0	0	0	0	0	0	0	0x0			
Access	RO	RO	RO	RO	RO	RO	RO	RO	RW			
Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	MCAST_ADDR [11:0]											
Default	0x00											
Access	RW											

MCAST_ADDR [15:0]	Multicast group address. In case that the device address indicates the multicast group address. The device can be assigned to multiple of the 16 multicast groups. All devices received the command act like a broadcast and all device belonging to a multicast address should treat the command transaction. This register can be accessed by SET_MCAST and READ_MCAST command. [0] = 1 : Multicast group address 0xFEf [1] = 1 : Multicast group address 0xFF0 *** [15] = 1 : Multicast group address 0xFFE 12 bits device addr 0x000 → broad-cast addr 0x001~0xFEE → individual node addr 0xFEf~0xFFE → multi-cast addr 0xFFF → default addr
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5.6.2.4 Setting of 8bits RGB Brightness controller

SET_RGB / READ_RGB :

Write / Return the RGB register

Bit	[23]	[22]	[21]	[20]	[19]	[18]	[17]	[16]	[15]	[14]	[13]	[12]
Name	R_DATA [7:0]								G_DATA [7:4]			
Default	0x00								0x0			
Access	RW								RW			
Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	G_DATA [3:0]				B_DATA [7:0]							
Default	0x0				0x00							
Access	RW				RW							
R_DATA [7:0]		Red color data is converted into 12-bit PWM data toward a sink driver. R_PWM data = (R_DATA x PWMMAX*) + 128) >> 8										
G_DATA [7:0]		Green color data is converted into 12-bit PWM data toward a sink driver. G_PWM data = (G_DATA x PWMMAX*) + 128) >> 8										
B_DATA [7:0]		Blue color data is converted into 12-bit PWM data toward a sink driver. B_PWM data = (B_DATA x PWMMAX*) + 128) >> 8										

(*) PWMMAX : Data stored on OTP

5.6.2.5 Setting of Over-Temperature Threshold

SET_TEMPTH / READ_TEMPTH ::

Write / Return the TEMPTH register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	RSVD	RSVD	TEMP_THRESHOLD[9:0]									
Default	0	0	0x000									
Access	RO	RO	RW									
TEMP_THRESHOLD [9:0]		Overtemperature fault detection threshold.										
		The OT_FLT is raised automatically the temperature increases above this value.										
		If TEMP_THRESHOLD is set to 0x000, the OT diagnosis is disabled.										
		To use the OT monitoring function, TEMP_THRESHOLD[9:0] value other than 0 must be set.										
		$TEMP_THRESHOLD[9:0] = (T_{J_OT} - 30^{\circ}C) \times 4.05 + d512$										

SET_TEMPHEYS / READ_TEMPHEYS :

Write / Return the TEMPHEYS register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	RSVD	RSVD	TEMP_HYSTERISIS[9:0]									
Default	0	0	0x000									
Access	RO	RO	RW									
TEMP_HYSTERISIS [9:0]		Overtemperature fault release threshold.										
		The OT_FLT can be cleared only when the temperature is below this value.										
		The TEMP_HYSTERISIS setting value must be set smaller than TEMP_THRESHOLD.										
		Otherwise, OT diagnosis will not work.										

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5.6.2.6 Setting of RGB MAX Current Level

SET_CURR_MAX_LVL / READ_CURR_MAX_LVL:

Write / Return the CURR_MAX_LVL register

Bit	[23]	[22]	[21]	[20]	[19]	[18]	[17]	[16]	[15]	[14]	[13]	[12]
Name	R_CURR_MAX_LVL[3:0]				G_CURR_MAX_LVL[3:0]				B_CURR_MAX_LVL[3:0]			
Default	RW				RW				RW			
Access	0x8				0x8				0x8			

This determines the max current value of the output each (R/G/B) sink driver.

- 4'b0000: 1.44mA
- 4'b0001: 2.87mA
- 4'b0010: 4.31mA
- 4'b0011: 5.75mA
- 4'b0100: 7.18mA
- 4'b0101: 8.62mA (B_CURR_MAX Recommended)
- 4'b0110: 10.05mA
- 4'b0111: 10.48mA
- R/G/B_CURR_MAX_LVL[3:0] 4'b1000: 12.92mA
- 4'b1001: 14.35mA
- 4'b1010: 15.78mA (G_CURR_MAX Recommended)
- 4'b1011: 17.21mA (R_CURR_MAX Recommended)
- 4'b1100: 18.65mA
- 4'b1101: 20.08mA
- 4'b1110: 21.51mA
- 4'b1111: 22.94mA

*The RGB LED chip integrated into this module is optimized to emit D65 White at 0xBA5.

5.6.2.7 Setting of Communication Time-out Length

SET_TIMEOUT / READ_TIMEOUT:

Write / Return the TIMEOUT register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	TIMEOUT [11:0]											
Default	0x800											
Access	RW											

TIMEOUT [11:0] Communication timeout length set.
TIMEOUT 1LSB : mclk(62.5ns) * 2^10 = 64us
Default TIMEOUT = (0x800)*(64us) = 131ms

5.6.3 Device Read command

5.6.3.1 READ STATUS1

READ_STATUS1 :

Return the STATUS1 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	RSVD	RSVD	RSVD	RSVD	RSVD	IC_STATE[2:0]			SIO2[1:0]		SIO1[1:0]	
Default	0	0	0	0	0	0x0			0x0		0x0	
Access	RO	RO	RO	RO	RO	RO			RO		RO	
IC_STATE[2:0]		Represent current IC state. 3'b000 : POR. (default) 3'b001 : Uninitialized. 3'b010 : Sleep. 3'b011 : Active. 3'b100 : Deepsleep.										
SIO2[1:0]		Communication mode of SIO2. 2'b00 : LVDS. (default) 2'b01 : EOL. 2'b10 : MCU 2'b11 : CAN Value is set automatically at startup and after reset.										
SIO1[1:0]		Communication mode of SIO1. 2'b00 : LVDS. (default) 2'b01 : EOL. 2'b10 : MCU 2'b11 : CAN Value is set automatically at startup and after reset.										

5.6.3.2 READ STATUS2

READ_STATUS2 :

Return the STATUS2 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	OTPCRC	T_OUT	OPEN_R	OPEN_G	OPEN_B	SHORT_R	SHORT_G	SHORT_B	OT_FLT	UV_FLT	CRC_FLT	COM_FLT
Default	0	0	0	0	0	0	0	0	0	0	0	0
Access	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC
OTPCRC		OTP error flag is set during startup when a mismatch between the OTP values and the internal CRC is detected. Cleared after readout										

T_OUT	Frame Time Out event flag is raised if the response is not arrived within a fixed time, and auto-cleared after readout.
OPEN_R	Red LED Open event flag is raised if $VLED,RED < VTH_OPEN$ for more than $Tdiag_filter$ time and the PWM output signal is active. If the event flag is raised under OS_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.
OPEN_G	Green LED Open event flag is raised if $VLED,GREEN < VTH_OPEN$ for more than $Tdiag_filter$ time and the PWM output signal is active. If the event flag is raised under OS_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.
OPEN_B	Blue LED Open event flag is raised if $VLED,BLUE < VTH_OPEN$ for more than $Tdiag_filter$ time and the PWM output signal is active. If the event flag is raised under OS_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.
SHORT_R	Red LED Short event flag is raised if $VLED,RED > VSHORT_TH$ for more than $Tdiag_filter$ time and the PWM output signal is active. If the event flag is raised under OS_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.
SHORT_G	Green LED Short event flag is raised if $VLED,GREEN > VSHORT_TH$ for more than $Tdiag_filter$ time and the PWM output signal is active. If the event flag is raised under OS_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.
SHORT_B	Blue LED Short event flag is raised if $VLED,BLUE > VSHORT_TH$ for more than $Tdiag_filter$ time and the PWM output signal is active. If the event flag is raised under OS_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.
OT_FLT	Over temperature fault flag is raised if $TEMPERATURE[9:0] > TEMP_THRESHOLD[9:0]$ for more than 5 consecutive $EVENT_CYC$ cycles time. If the event flag is raised under OT_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout in $TEMPERATURE[9:0] < TEMP_HYSTERISIS[9:0]$.
UV_FLT	Undervoltage fault flag is raised if $VDD < Vuvth_lo$ for more than 5 consecutive $EVENT_CYC$ cycles time. If the event flag is raised under UV_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.
CRC_FLT	CRC fault flag is raised if CRC is wrong and auto-cleared after readout.
COM_FLT	Communication fault flag is raised if any communication error has occurred, and auto-cleared after readout. If the fault flag is raised under COM_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off.

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5.6.3.3 READ DIE_TEMP

READ_TEMP :

Return the TEMP register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	RSVD	RSVD	TEMPERATURE[9:0]									
Default	0	0	0x000									
Access	RO	RO	RO									
TEMPERATURE[9:0]		The die temperature sensor value is converted through 10bit ADC.										

5.6.3.4 READ DIE_TEMP and STATUS2

READ_TEMPST :

Return the TEMP & STATUS2 register

5.6.3.5 READ Calculated PWM value for R/G/B channel

READ_PWM_RED_VAL :

Return the PWM_RED_VAL register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	PWM_RED_VAL[11:0]											
Default	0x0000											
Access	RO											
PWM_RED_VAL [11:0]		12bit RED PWM calculated data value										

READ_PWM_GREEN_VAL :

Return the PWM_GREEN_VAL register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	PWM_GREEN_VAL[11:0]											
Default	0x0000											
Access	RO											
PWM_GREEN_VAL[11:0]		12bit GREEN PWM calculated data value										

READ_PWM_BLUE_VAL :

Return the PWM_BLUE_VAL register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	PWM_BLUE_VAL[11:0]											
Default	0x0000											
Access	RO											
PWM_BLUE_VAL[11:0]		12bit BLUE PWM calculated data value										

6. Typical Electrical Optical Characteristics Curves

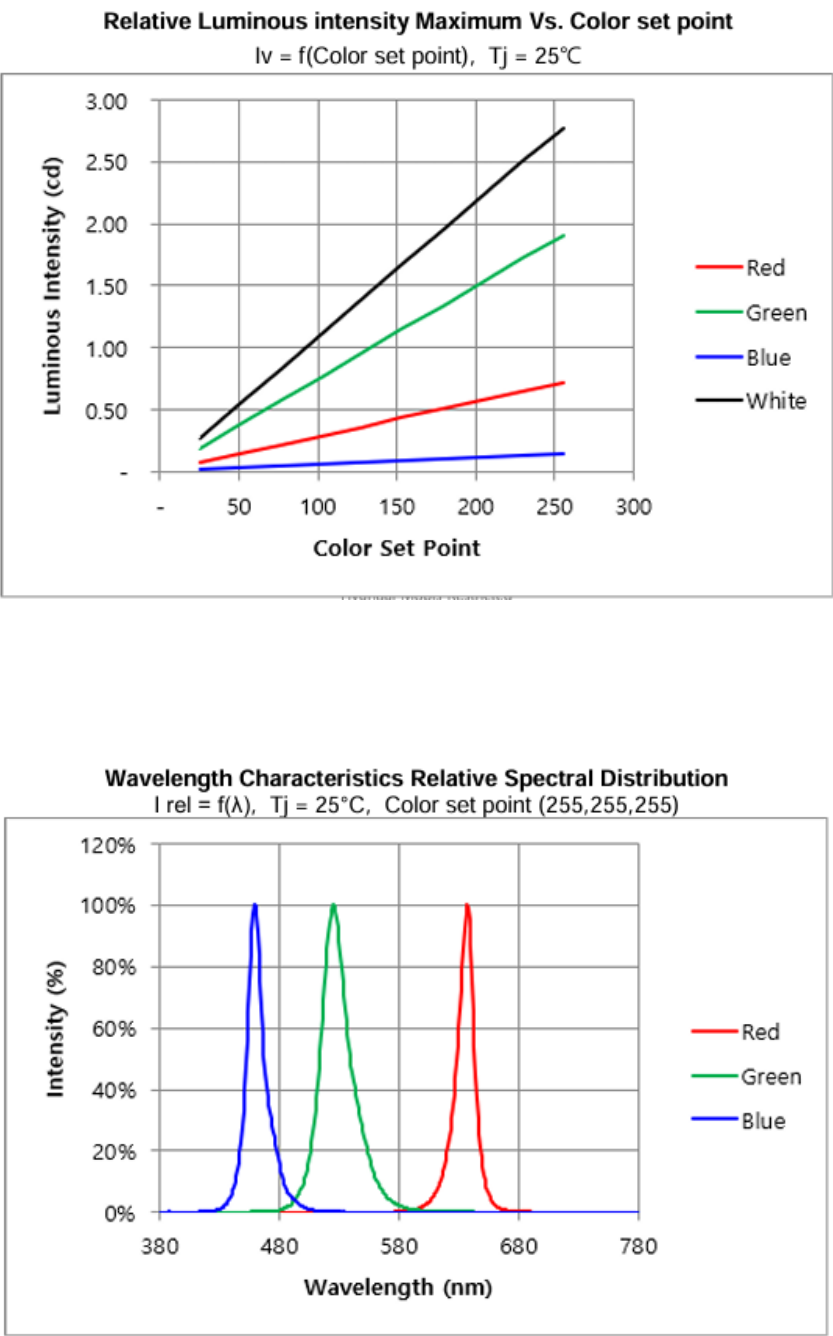


Fig 17. Typical Electrical Optical Characteristics Curves

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7. Soldering Temperature Reflow Profile

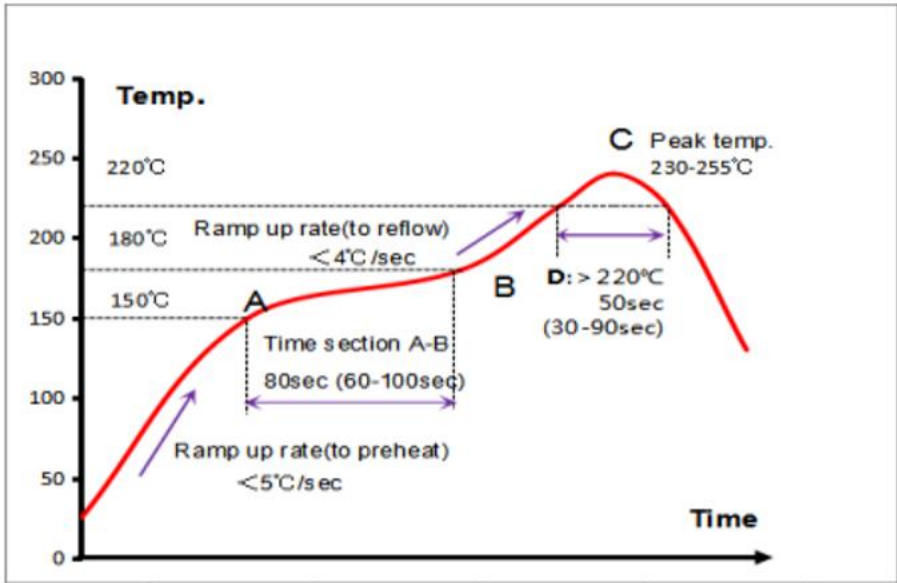


Fig 18. Reflow Soldering information

Point		Min.	Typ.	Max.	Unit
A	Pre-heat start point	140	150	160	°C
B	Pre-heat end point	160	180	200	°C
A-B	Pre-heat time	60	80	100	s
C	Peak temp.	230	240	255	°C
D	Time above 220°C	30	50	90	s

Table 14. Reflow Soldering Parameter

Note:

1. The reflow peak soldering temperature is peak 255°C and it is specified based on IPC/JEDEC J-STD- 020 “Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices”
2. Tolerance for peak profile temperature (Tp) is defined as a supplier minimum and a user maximum.
3. This document should serve as recommendation only. Other parameters may also affect soldering, so these profiles do not guarantee absolute success.
4. Soldering profiles should be determined by reference to the solder and solder paste manufacturer's specifications, unless they conflict with the recommendations in this document.
5. We do not provide explicit or implied guarantees for the accuracy of the data. We are not liable for any loss caused by the use of this information or specified materials
6. The soldering profile and footprint information specified in this datasheet are general recommendations and are provided for reference only. The design, control, and quality assurance of the actual soldering process are the sole responsibility of the customer. The supplier assumes no responsibility for the soldering process or its results.

Important statement

Global Technologies reserves the right to change the above-mentioned information without prior notice.

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8. Package Description

8.1 Package outline Dimension

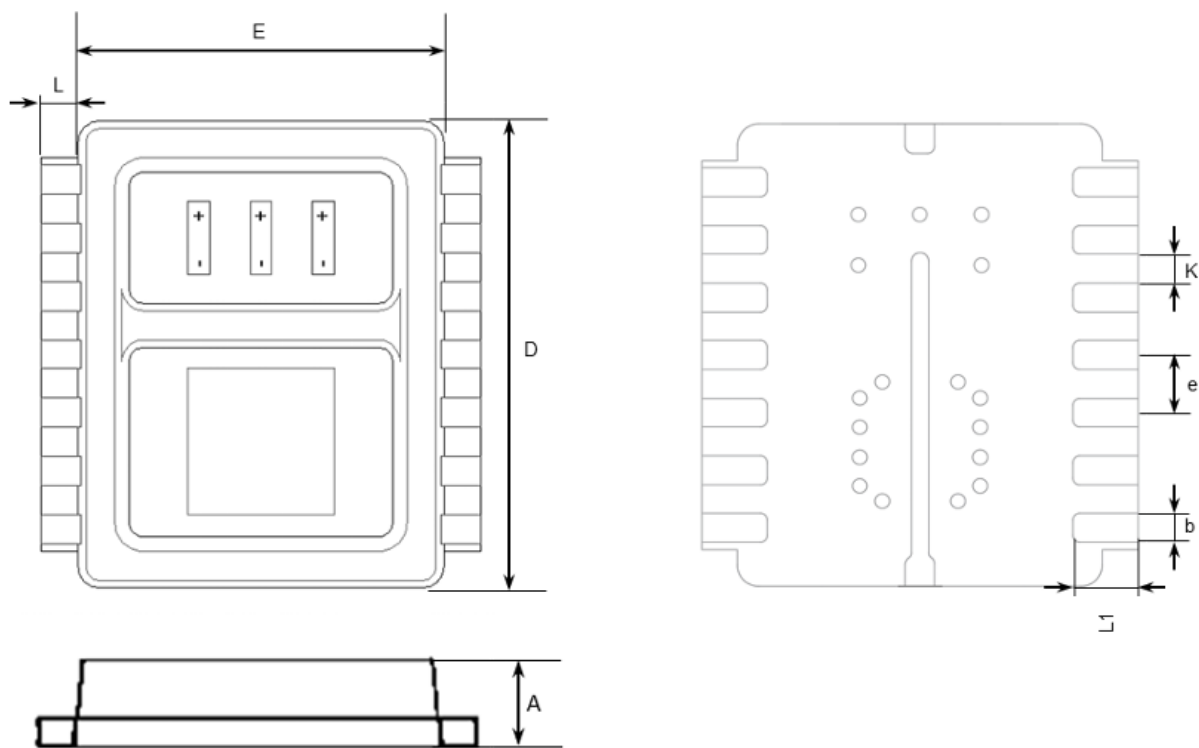


Fig 19. POD (DFN 3225-14L)

Symbol	Dimensions in mm		
	Min	Nom	Max
A		0.65	
b	0.20 BSC		
D		3.20	
E		2.50	
e	0.40 BSC		
K	0.20 BSC		
L		0.30	
L1		0.45	

General tolerance is $\pm 0.1\text{mm}$.
RGB LED Pitch is 420um.

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8.2 SAL IC Image

Package : DFN3225-14L

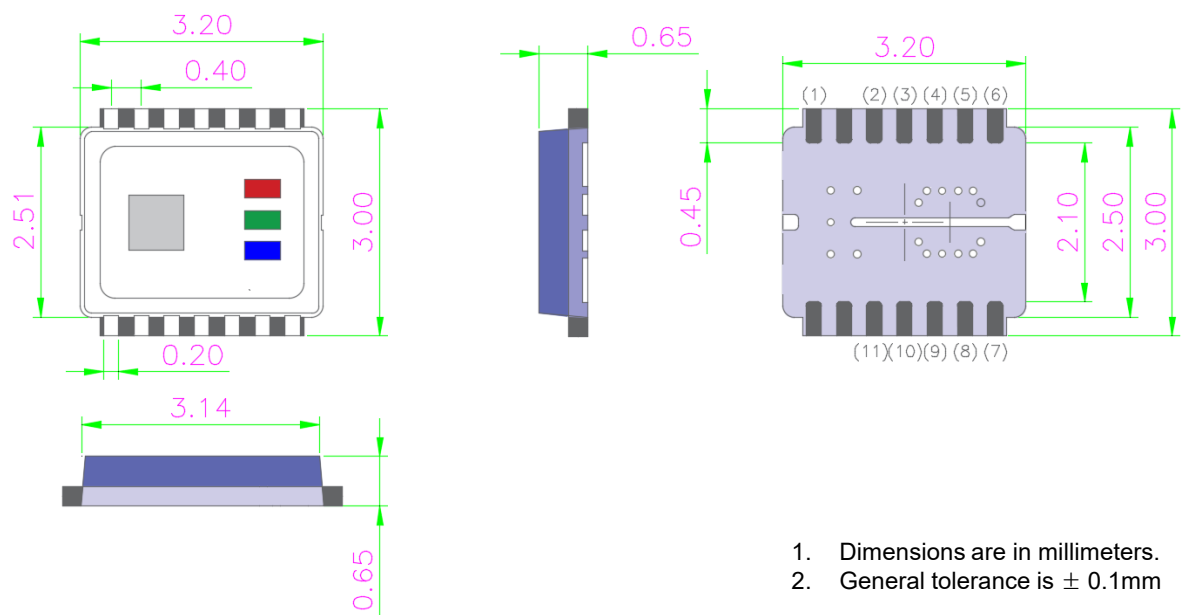


Fig 20. Packing outline Dimension

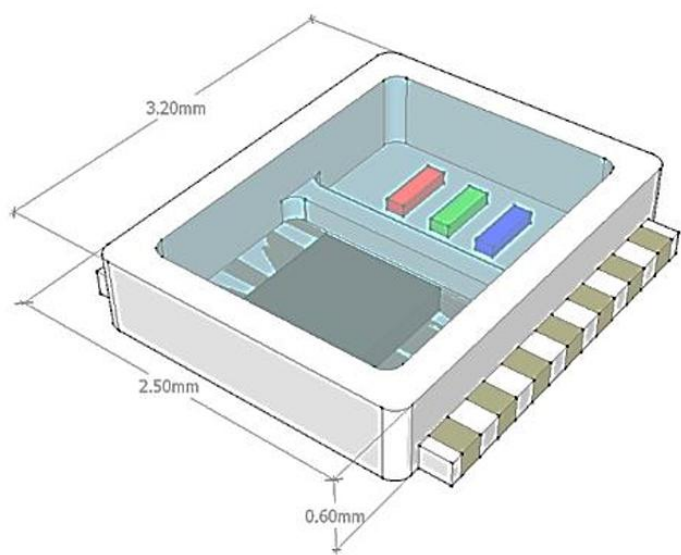


Fig 21. SAL IC Image diagram

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8.3 Recommended Solder Pad Layout

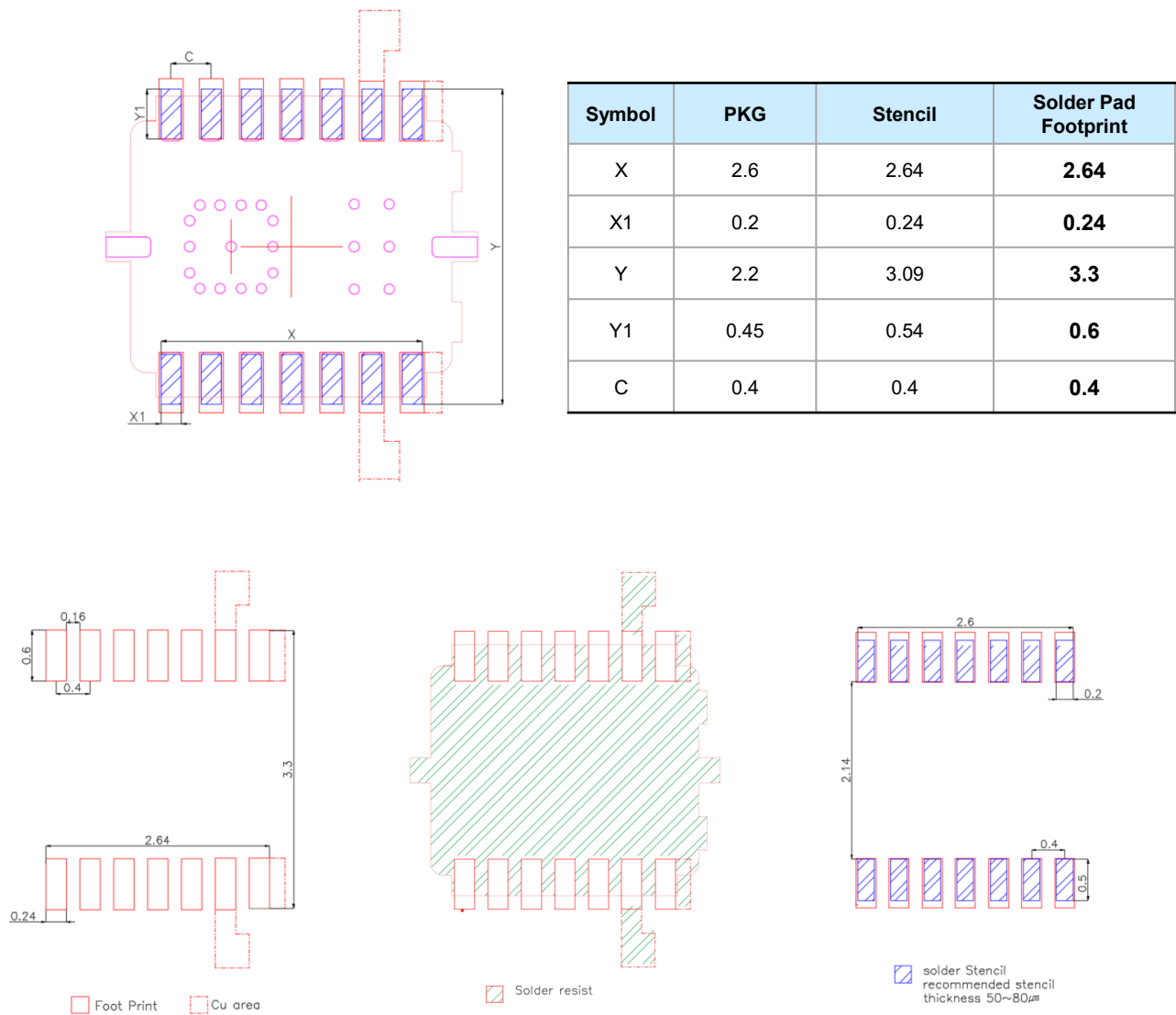


Fig 22. Solder Pad Layout

Note: The reflow peak soldering temperature is peak 255°C and it is specified based on IPC/JEDEC J-STD - 020 “Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices”

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9. Handling Precautions

9-1. Handling precautions

1.Storage

- It is recommended to store the products in the following conditions:
 - Humidity: 10% ~ 55% R.H.
 - Temperature : 18°C~28°C(64.4°F~82.4°F)
- Shelf life in sealed bag: 12 month at < 18°C~28°C and < 10% ~ 55% R.H.
- After the package is Opened, the products should be used within 1 weeks or they should be stored in a vacuumed and sealed with silica gel inside at 30°C or less and 20% R.H. or less.
- After opening the package, the LEDs should be kept at 30°C or less and 20%RH or less.
- When restoring unused the LEDs with anti-electrostatic bag, seal off the anti-electrostatic bag so that no gas and humidity can get it.
- When storage the LEDs in the corrugated cardboard, it may make the LEDs discolored because of minute sulfur gas from the corrugated cardboard.
- Do not use over 10 days in case of using corrugated cardboard.
- Recommend corrugated cardboard containing sulfur less than 850ppm when It is Inevitable use of corrugated cardboard.
- Recommend using material type of PP or PET tray to storage the PCBs or assemblies containing the LEDs. and Insert the silica gel into each of tray.
- Use the anti-electrostatic box with anti-electrostatic cover to prevent Volatile Organic Compounds, sulfur gas and humidity when storing the bundle of the PCBs or assemblies containing the LEDs.
- Do not use bubble wrap directly on top of LEDs. It may cause damage the LEDs
- Do not use rubber band.
- The LEDs must be stored in a clean environment.
- Do not expose the LEDs to direct sunlight.
- Before opening the package, the LEDs should be kept at 18°C~28°C or less and 10% ~ 55% RH or less.
- The LEDs should be used within a year.

2.Baking

- Suggest packing open after 1 weeks, before use baking products, conditions as follows:
 - ① $60\pm 3^{\circ}\text{C}$ X 6hrs and $< 5\%\text{RH}$, for reel .
 - ② $125\pm 3^{\circ}\text{C}$ X 2hrs, for single LED
- It shall be normal to see slight color fading of carrier (light yellow) after baking in process.

3.Heat Generation

- When the LEDs are illuminating, operating current should be decided after being considering the ambient maximum temperature.
- Please consider the heat generation of the LED when it is designed the PCB.

4.Cleaning

- If user needs cleaning of the LEDs, use of Isopropyl Alcohol or Ethylene Alcohol is recommended in 5 minutes at room temperature. and dry at room temperature for 15 minutes before use the LEDs. If user uses other than Isopropyl Alcohol or Ethylene Alcohol as cleaning material, it should be not dissolve the LEDs.
- Freon solvents should not be used to clean the LEDs because of worldwide regulations.

5.Static Electricity

- Static electricity or surge voltage damages the Power SMD . It is recommended that a wrist band or an anti-electrostatic glove be used when handling the LEDs.
- A tip soldering iron is requested to be grounded. An ionizer should also be installed where risk of static.
- All devices, equipment and machinery must be properly grounded (via $1\text{M}\Omega$).
- If the LEDs is applied at voltage over maximum value, it may cause damage or destruction of the LEDs
- The LEDs damaged or destructed by anything may cause an increase in leak current, lowered turn on forward voltage, or the LEDs at low forward current.
- It is important to eliminate the possibility of surge current when user designs circuit

6. Product Warranty and Quality Conditions

- The warranty period for this product is one (1) year from the date of shipment.
- Product manual versions may be modified without prior notification. Customers are responsible for obtaining the most recent version of information prior to ordering and must verify that the information is complete and up-to-date.
- All semiconductor products may fail or malfunction under specific conditions. When using this product, the purchaser must comply with safe usage standards and take necessary safety precautions to prevent potential risks that could cause personal injury or property damage.

9-2. Handling LED

- User's working and testing environment can be a important factor to discolor. because unclean testing chamber or working place with wet floor may cause discolor of the LEDs.
- When handling the LED with tools like tweezers or nipper, do not apply Mechanical Forces directly on LED's Surface.
- Do not directly touch LED's surface with hand . It may contaminate the surface and affect on optical characteristics.
- The LEDs should be handled from side because LED's molding material may be damaged with scratching on surface, piercing molding material and broking wire.
- When doing surface mounting technology, If the encapsulation material of the LEDs is silicone, do not any stress or pressure that may cause that emitting surface area of the LEDs resin can be cut, chipped, delaminate or deformed, causing wire bonding breaks and destruction of the LEDs.
- Recommended that the picking up nozzle optimize wider than emitting surface area of the LEDs and setting so that it does not damage the silicone resin.
- Maintain cleanness of picking up nozzle.
- Dropping the LEDs may cause damage.
- Do not contaminate emitting surface area of the LEDs.

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